

Physical Optics Sim - Documentation

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Purpose

The code implements fresnel and fraunhofer diffraction techniques, as well as beam steering to accurately incorporate all aspects of this process. The overall goal of this project is to replicate the mirror calibration process that is used on BICEP array so that we can compare the data from the telescope beam maps and the simulation, to find a quantitative assessment of how the mirror non-flatness might be affecting areas of the beam maps.

Global Variables

Global variables allow for customazation of the simulation. The variables are as follows:

Variable	What it does
N	number of grid points
Radius	Radius of circular aperture
dx	spacing between grid points
edge taper	controls the gaussian fade
theta	.
lamda	.
mirror coord	.
phase gradient	.

Table 1: Available variables

Aperture Creation

The aperture function focuses on creating a 2d spatial grid that defines a circular aperture. It uses an N number of grid points, with a dx spacing between each of the points.

The dx remains the same for the aperture function and the fresnel function, since both use the same physical grid spacing. (1)

The coordinate array is centered at zero and a gaussian taper is applied to replicate the illumination of the aperture in experimental values.

Mirror Creation

The mirror function takes the mirror coordinates provided in the global variables section and focuses on DISTORTION:

Fresnel Diffraction

This function uses fresnel teqniques as found in (ref Optical physics third edition) to calculated the diffraction pattern from the aperture onto the mirror. Rather than using an FFT here, a for loop is used with the diffraction

by Huygen's principle:

$$E(x', y') = \iint A(x, y) \frac{e^{ikd}}{d} dx dy$$

This ensures that the Fresnel diffraction technique is preserved and accurate in the near field.

ACCOMODATING THE ANGLED REFLECTION: (multiply by -1 terminology) Since the Fraunhofer calculation is simulated as it reflects off the mirror at a 45 degree angle, the Near field output needs to be multiplied by -1 to account for this change in direction. According to Euler's formula, we can multiply the following phase shift by the produced output to achieve this effect.

$$e^{i\pi} = \cos(\pi) + i\sin(\pi) = -1 + 0j = -1$$

BEAM STEERING

$$\phi = k(x\cos(\theta) + y\sin(\theta))\tan(\alpha)$$

In order to steer the beam source two different angles must be defined. Here, there is a tilt angle (

$$\alpha$$

) and a rotation angle (

$$\theta$$

) on the x-axis. (insert ref)

Fraunhofer Diffraction

When doing Fraunhofer diffraction from the plane of the mirror to the far field, an FFT can be used.

Peak Normalization techniques are used to better view the diffraction patterns. Both a log scale and the linear scale of the peak normalization are visible below:

In order to keep the beam steering, and tilts applied to the aperture and mirror, the nearfield output from the fresnel diffraction function is input into the fraunhofer function. This allows for all features applied to be carried through accurately into the far field projection.

Global scaling for FFT:

$$\Delta f_x = \frac{1}{N \cdot \Delta x}$$

The output grid of an FFT is typically in meters (-1), these spatial frequencies map to physical angles in radians using the light's wavelength:

$$\Delta \theta_x = \frac{\lambda}{N \cdot \Delta x}$$

On going questions

References

- Peak Normalization - Fraunhofer function: <https://blogs.bath.ac.uk/python/finding-maximum-values-from-arrays/>
- Diffraction Scaling - Voelz, David George, and Society of Photo-optical Instrumentation Engineers. Computational Fourier Optics a MATLAB Tutorial. SPIE, 2011, <https://doi.org/10.1117/3.858456>.
- Goodman, Joseph W. Introduction to Fourier Optics. McGraw-Hill, 1968.